Supporting_MT

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2025-01-08

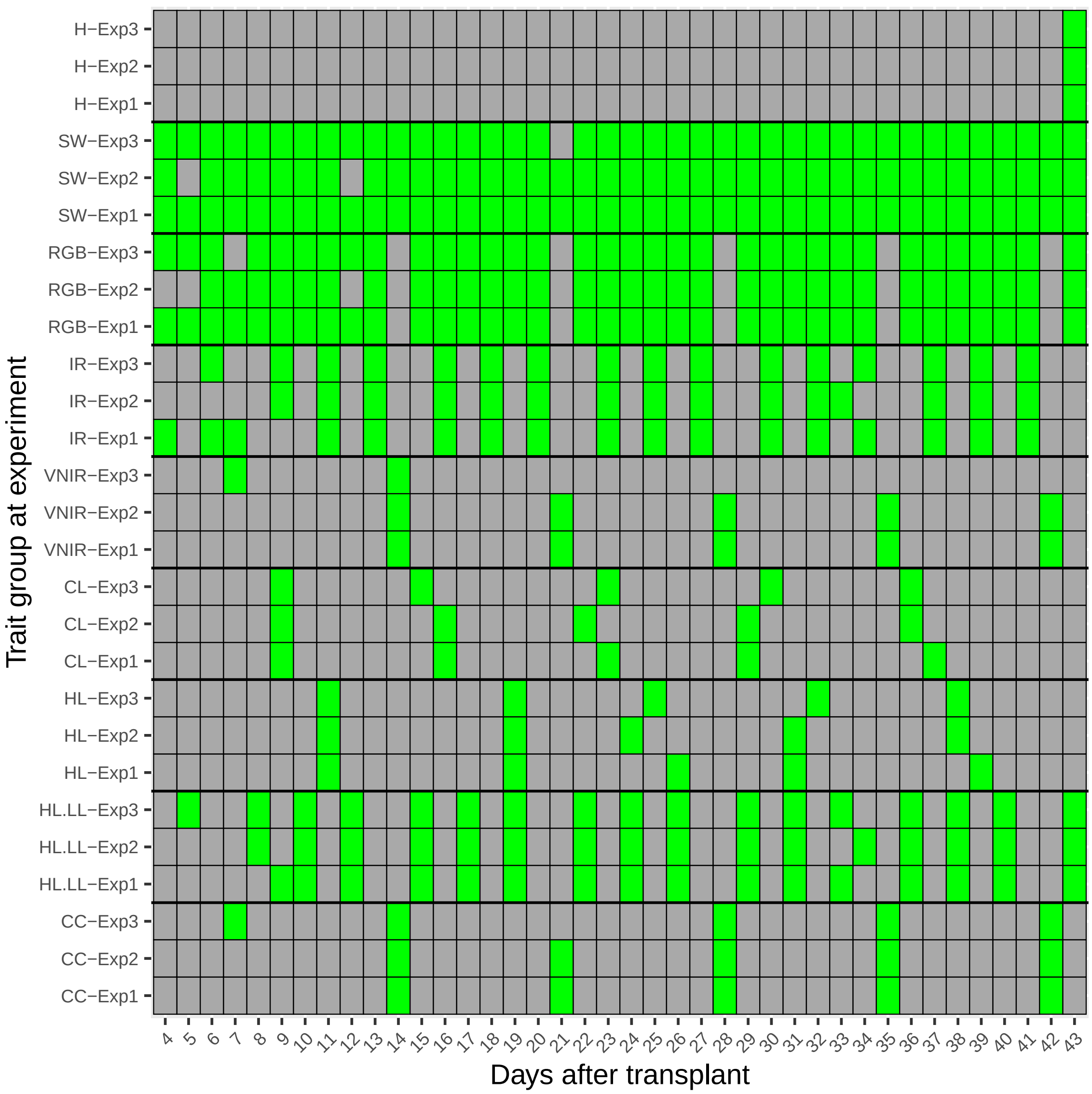

Figure 1: Experiment-wise measurement log for each group. The y-axis labels consist of a group prefix and an experiment suffix, denoting the group-experiment pair. A green rectangle indicates that a measurement protocol was conducted.

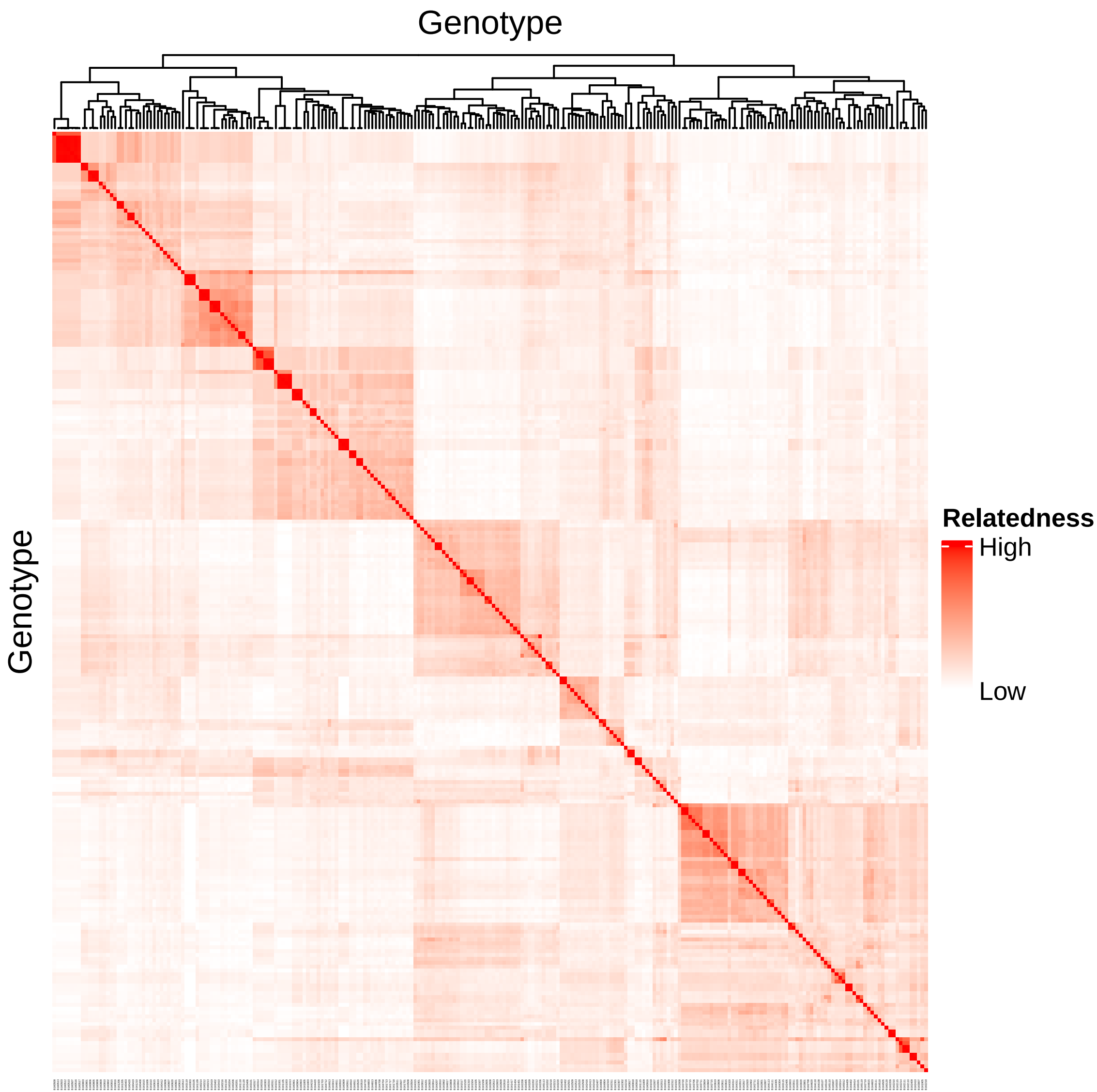

Figure 2: Kinship between 340 B1K accessions Relatedness is indicated in red. Above the heatmap columns, a dendrogram of a hierarchical clustering is plotted. The clustering is calculated using hclust in r.

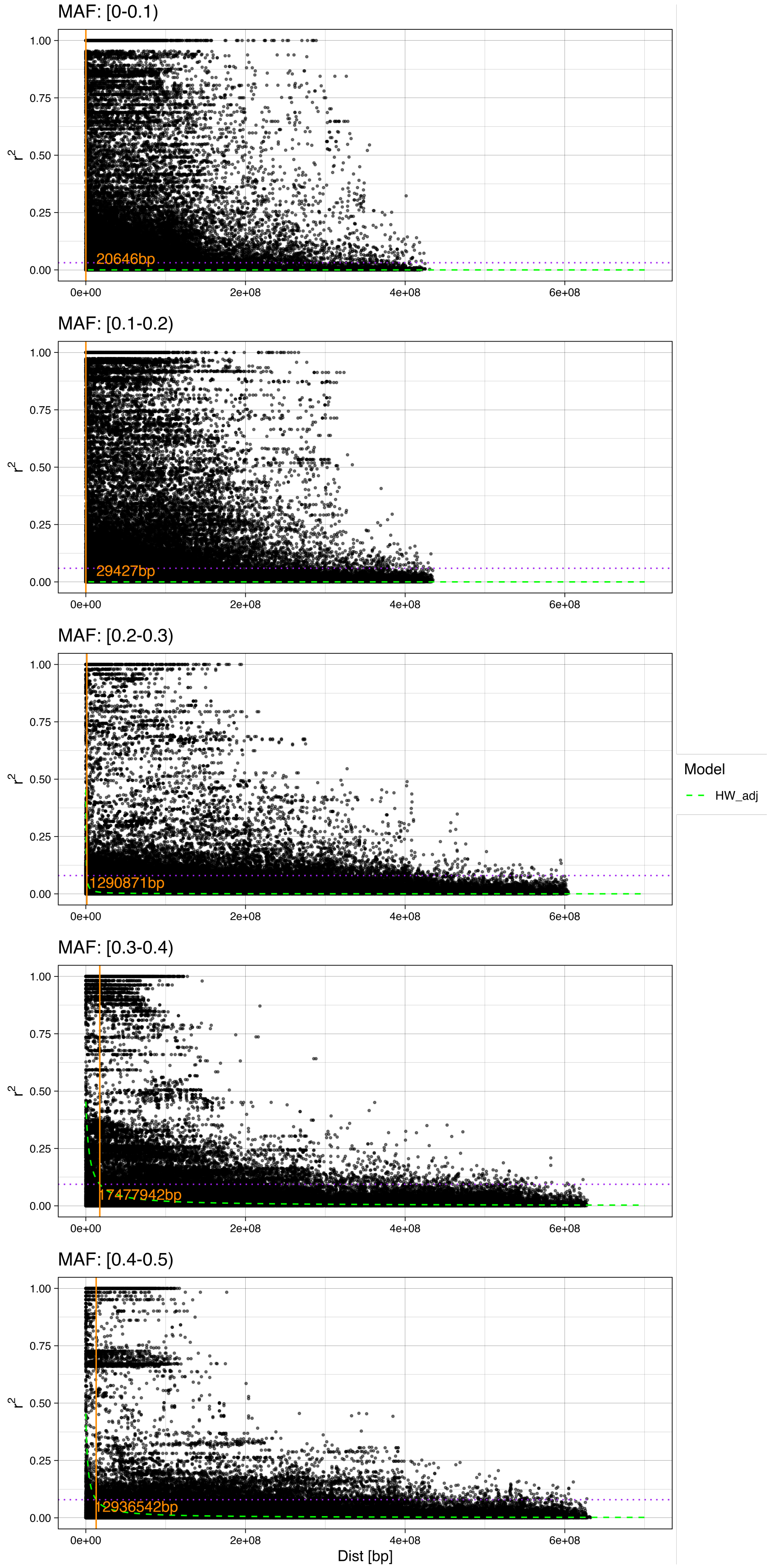

Figure 3: LD for maker pairs on five different MAF windows. The dotted magenta line indicates the background LD as defined in Materials and Methods. Best fits for Hill and Weir model are shown as dashed lines and plotted for 100000 distance values equally spaced on the x-axis.

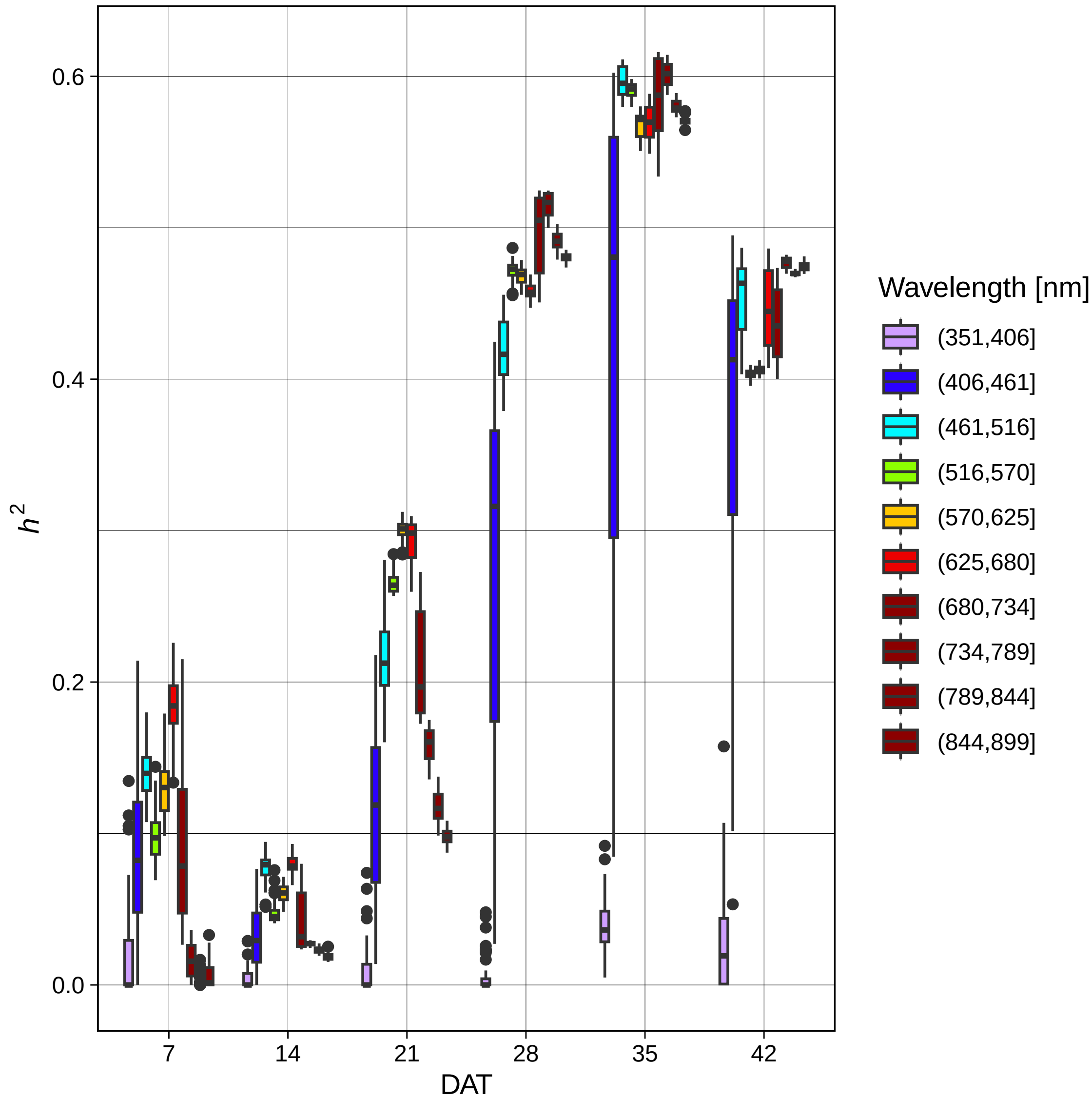

Figure 4: Heritability estimates for HSR-data for six measured time points. The measured wavelenghts are binned and color coded based on their length.

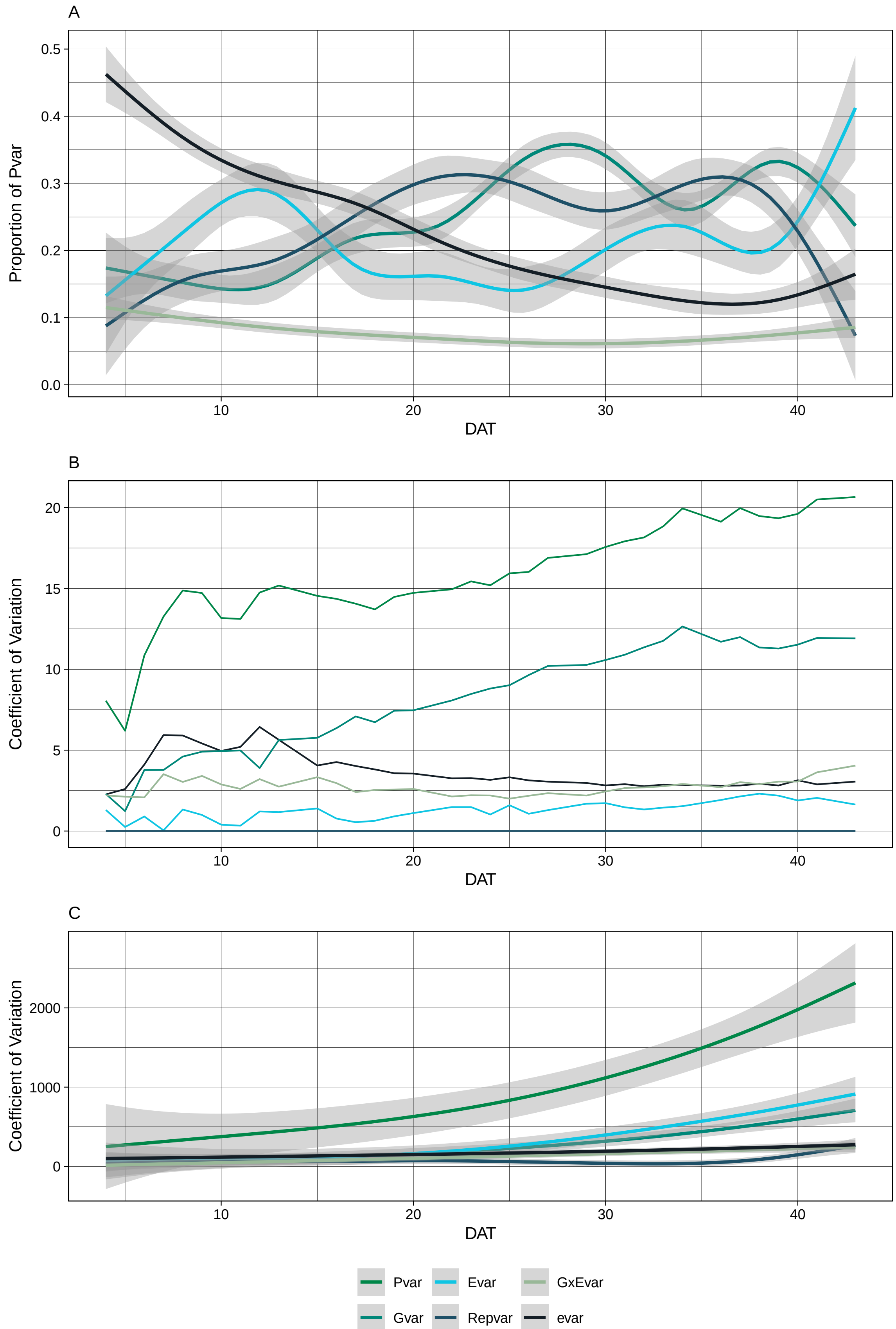

Figure 5: Relative contribution of variance components to phenotypic variance (a) The estimates are averaged over all models evaluated at each time point. (b) The coefficient of variation is calcuculated for every considered source of variation, using the mean phenotypic value at each timepoint for PH (RGB1_Plant_Avg_HEIGHT_MM). (c) The values calculated in (b) averaged over all traits.

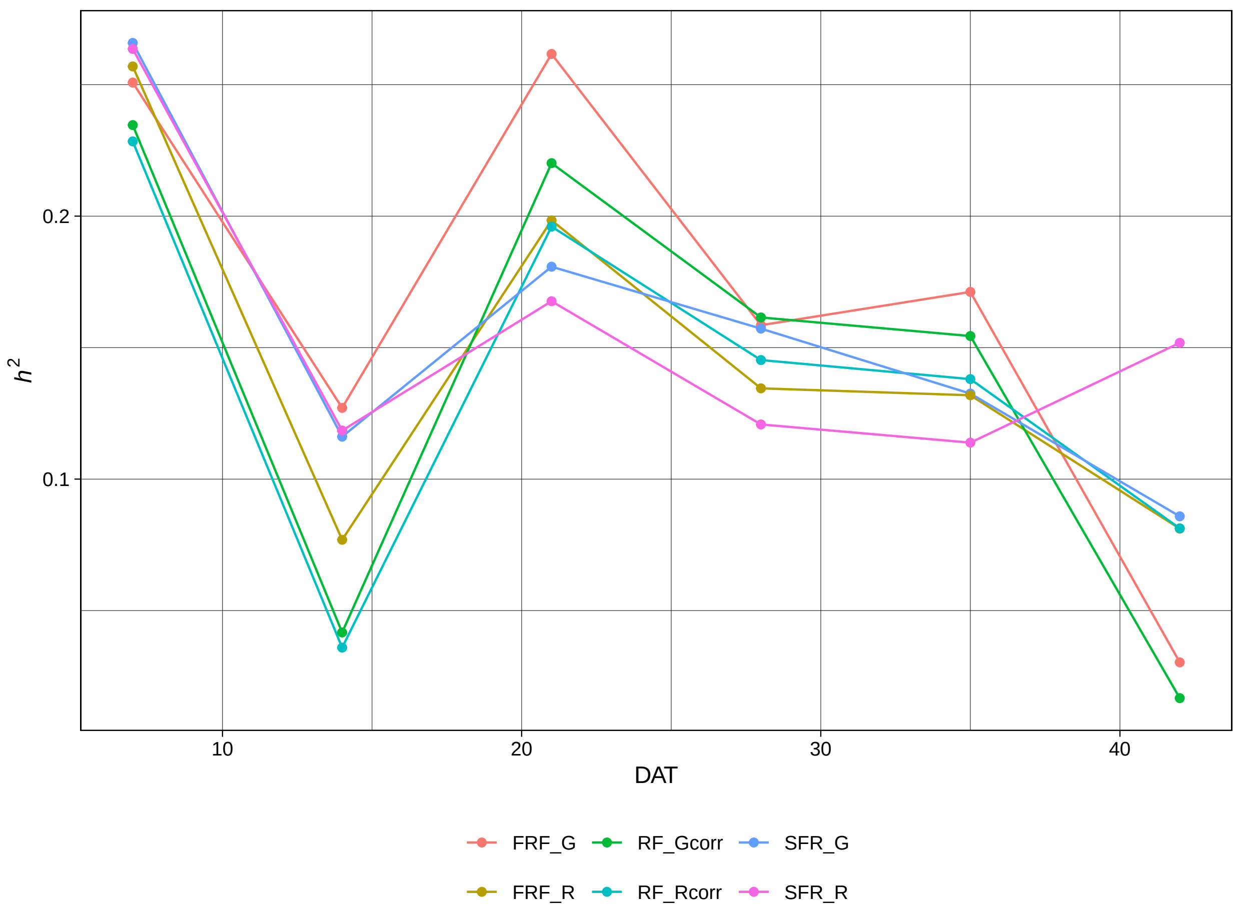

Figure 6: Heritability estimates for CC traits through time.

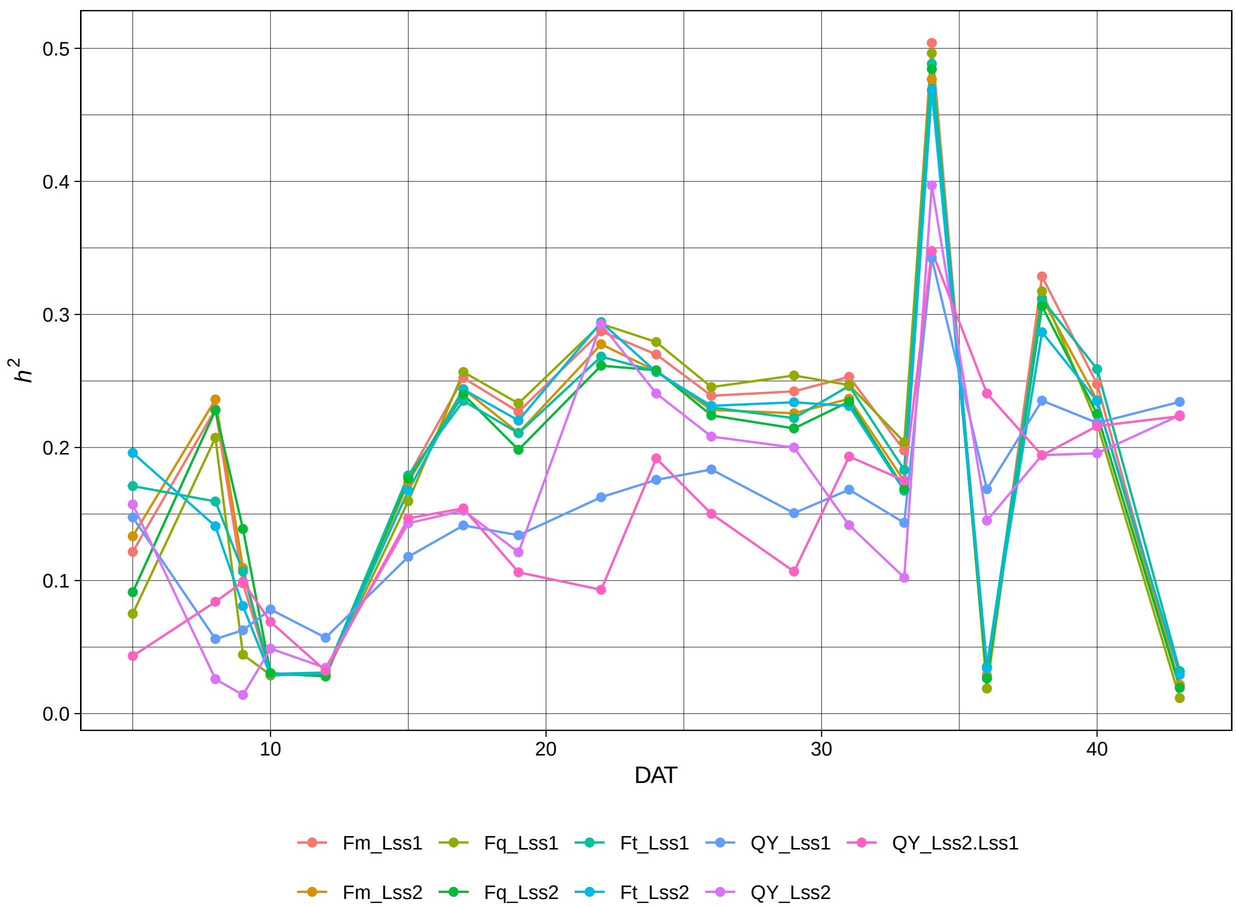

Figure 7: Heritability estimates for HL.LL traits through time.

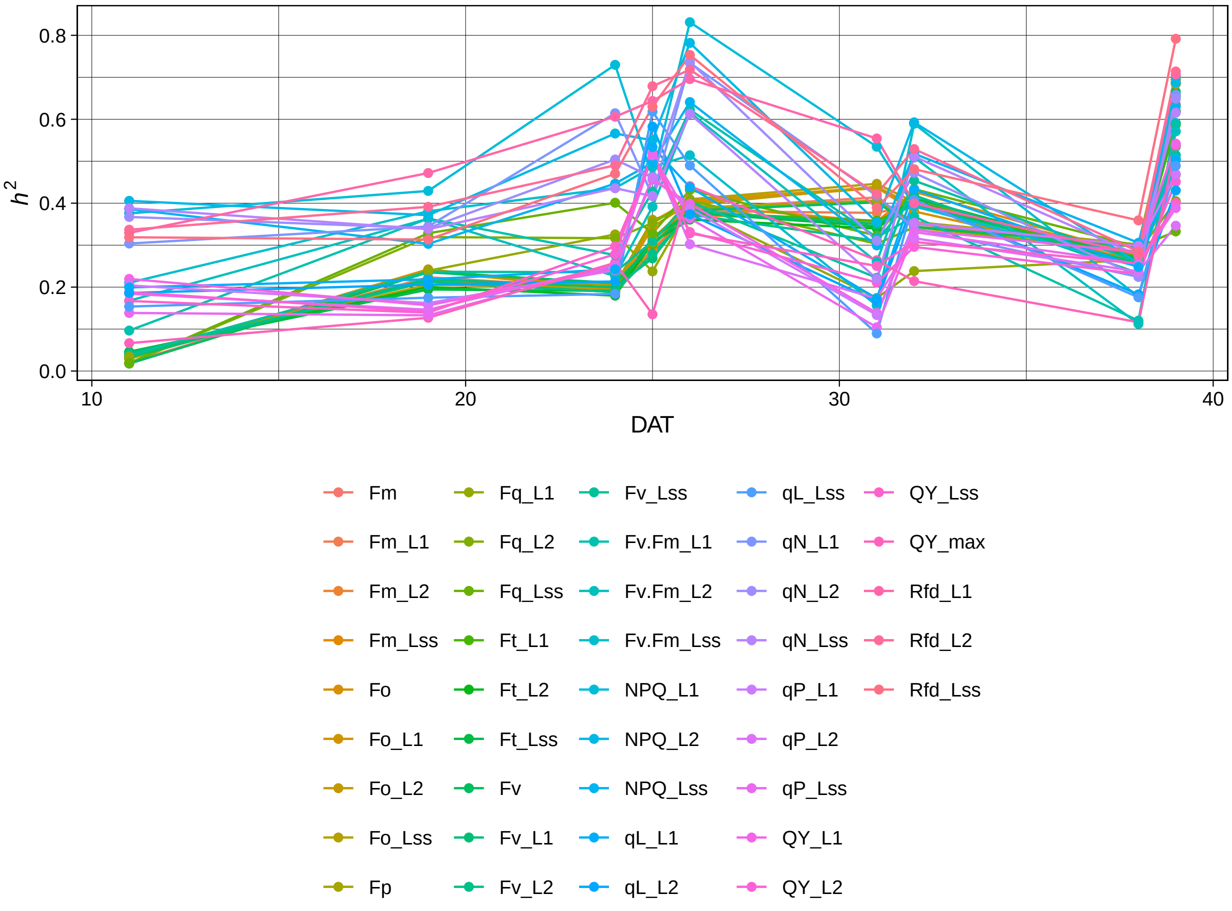

Figure 8: Heritability estimates for HL traits through time.

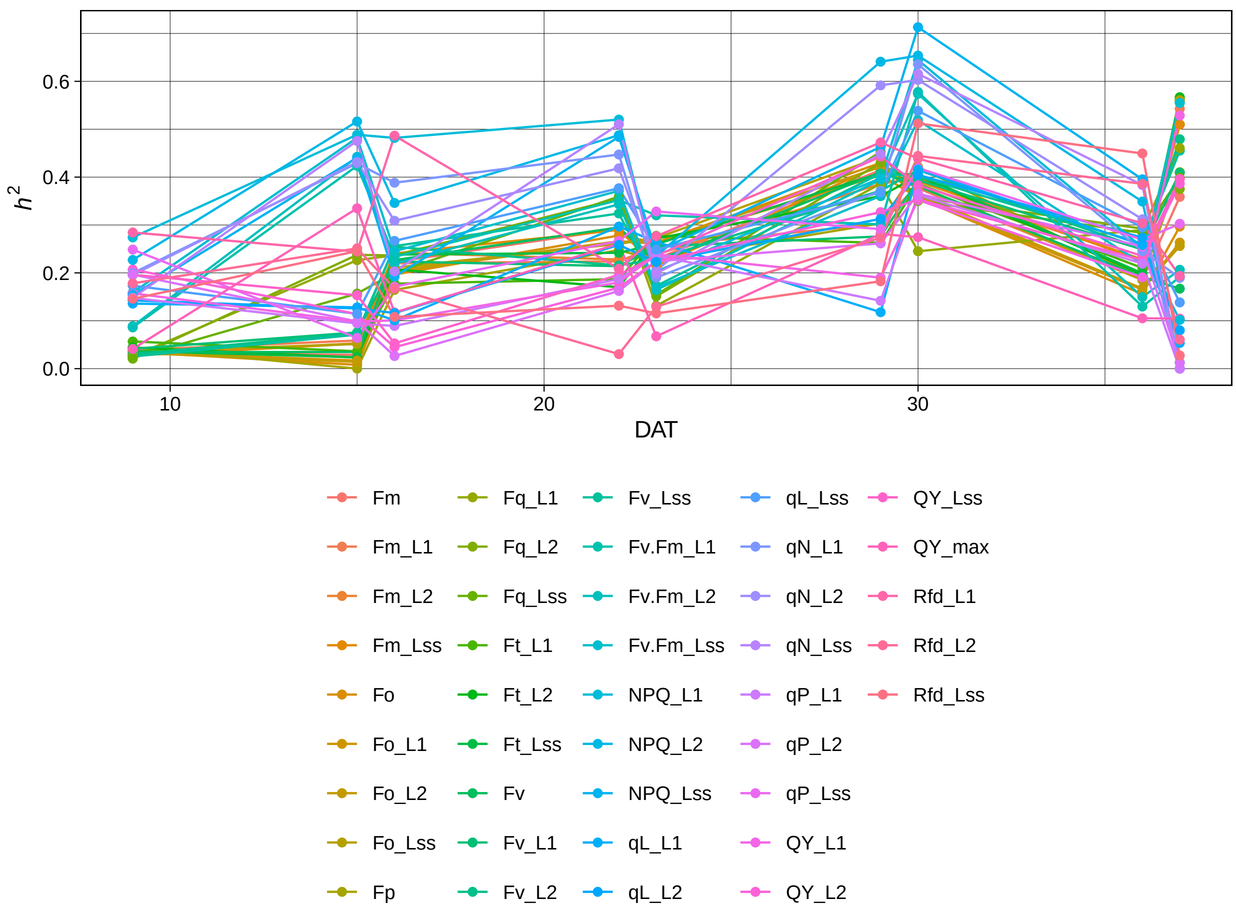

Figure 9: Heritability estimates for CL traits through time.

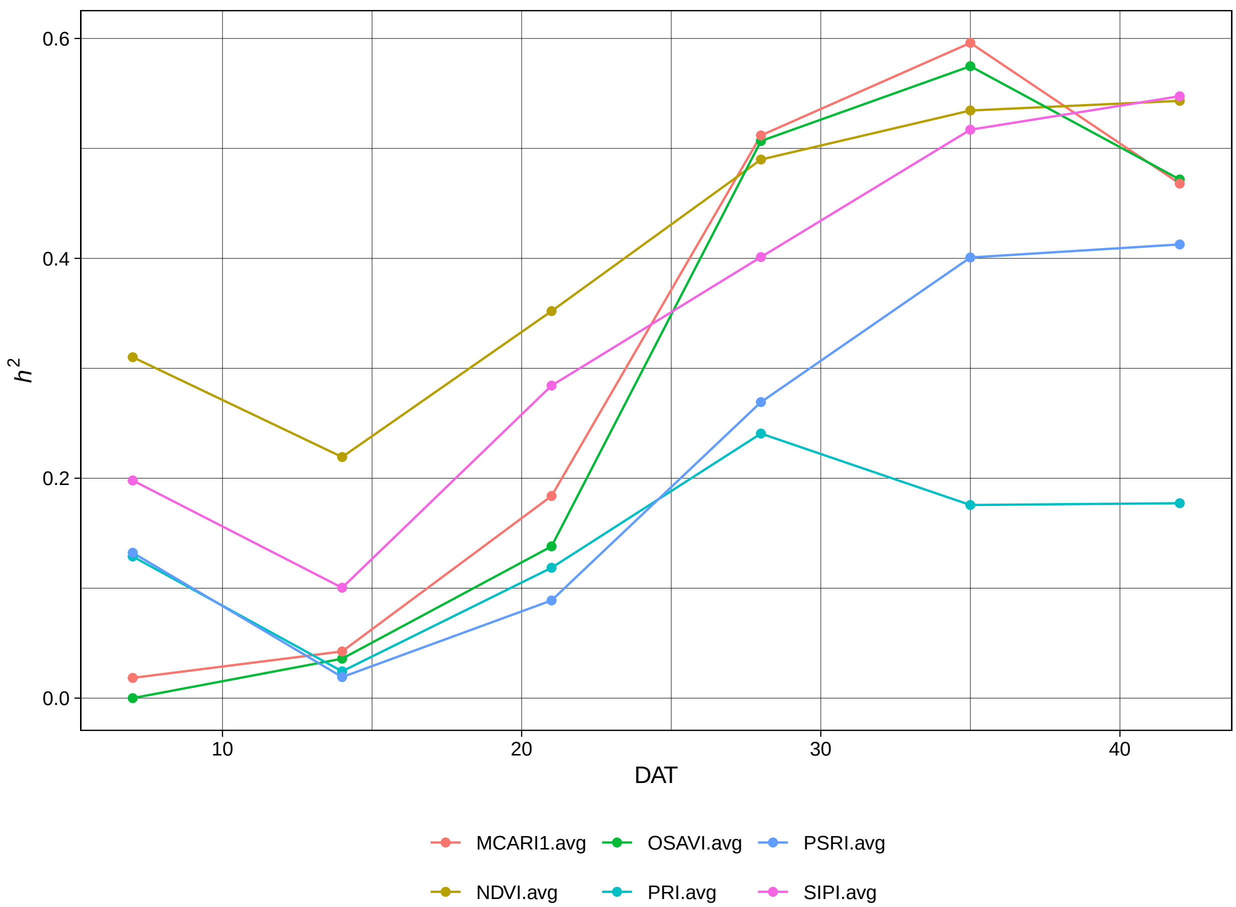

Figure 10: Heritability estimates for VNIR traits through time.

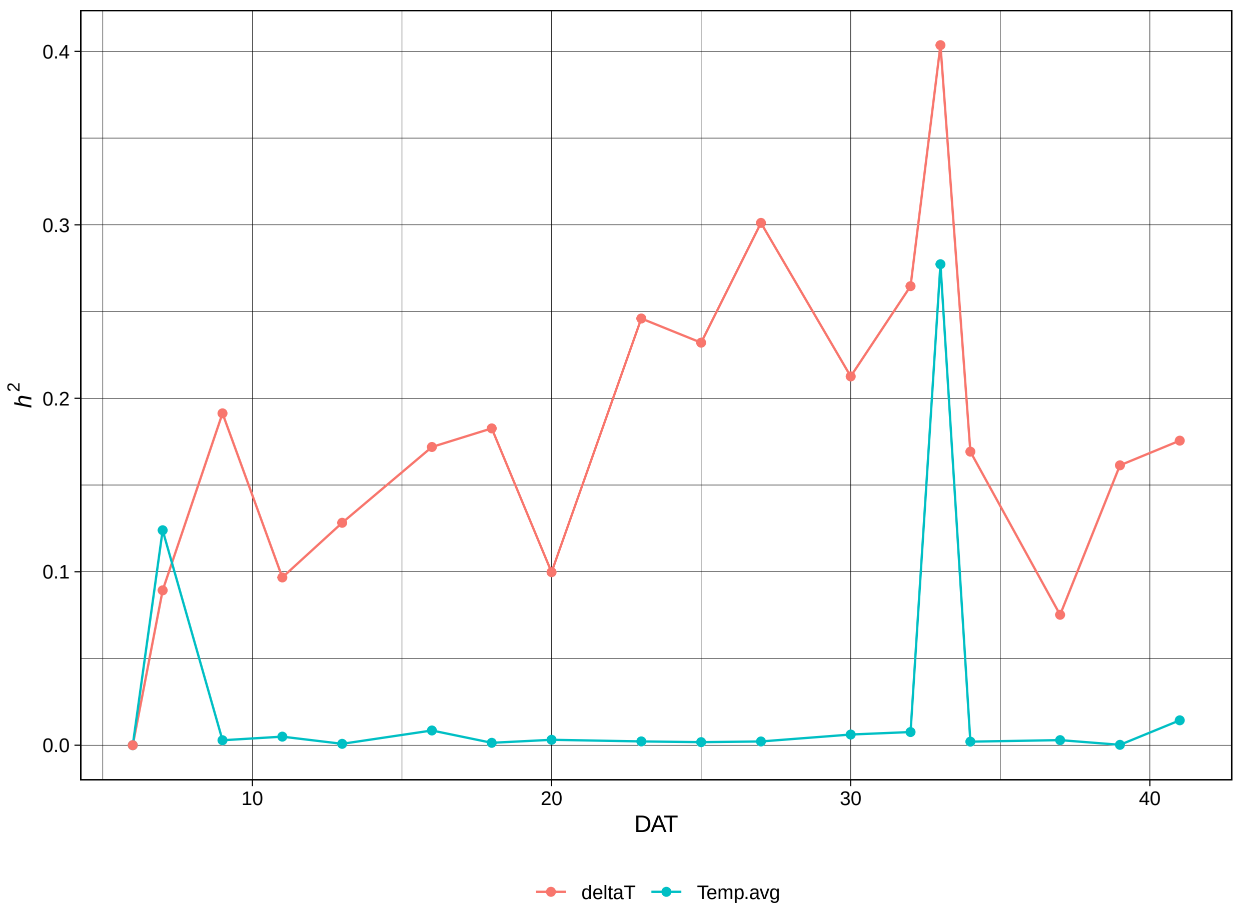

Figure 11: Heritability estimates for IR traits through time.

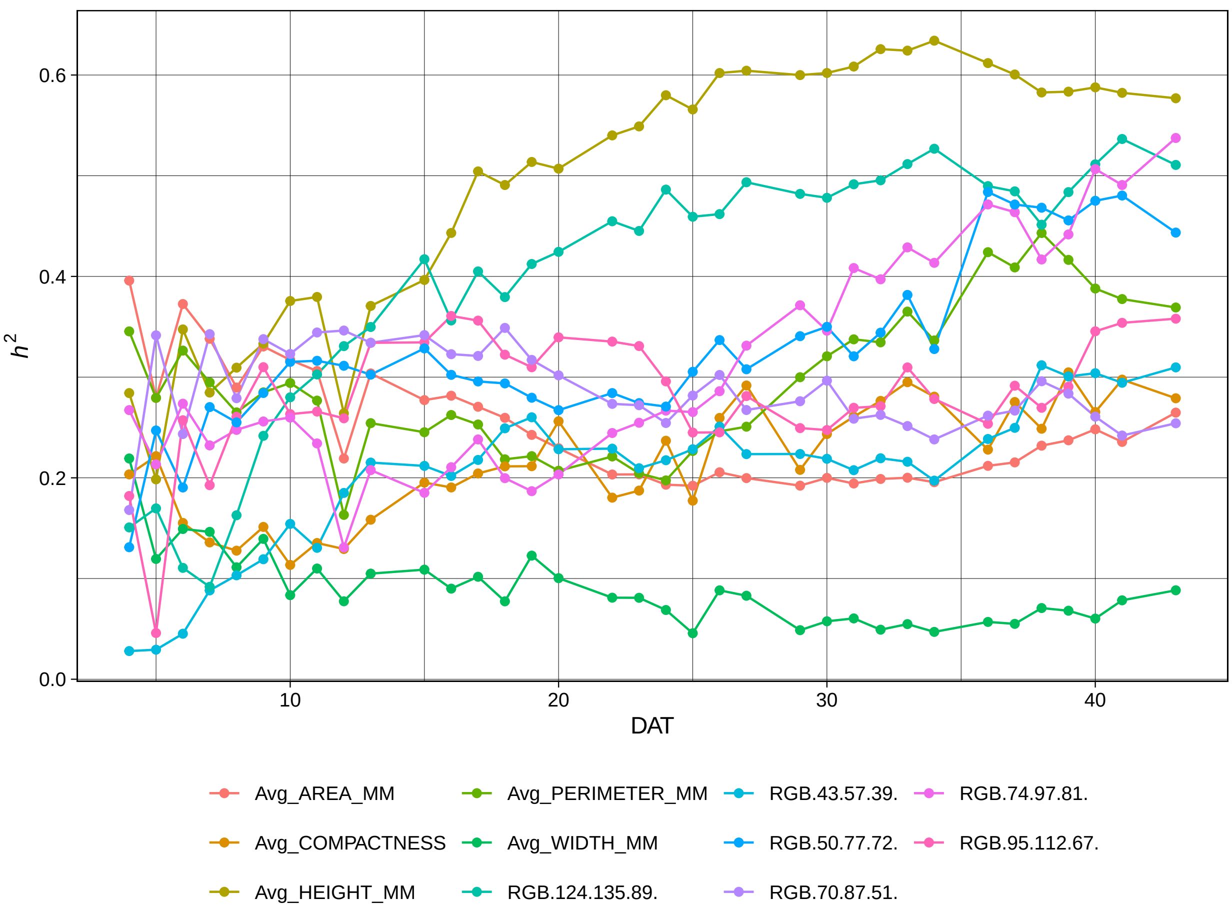

Figure 12: Heritability estimates for RGB traits through time.

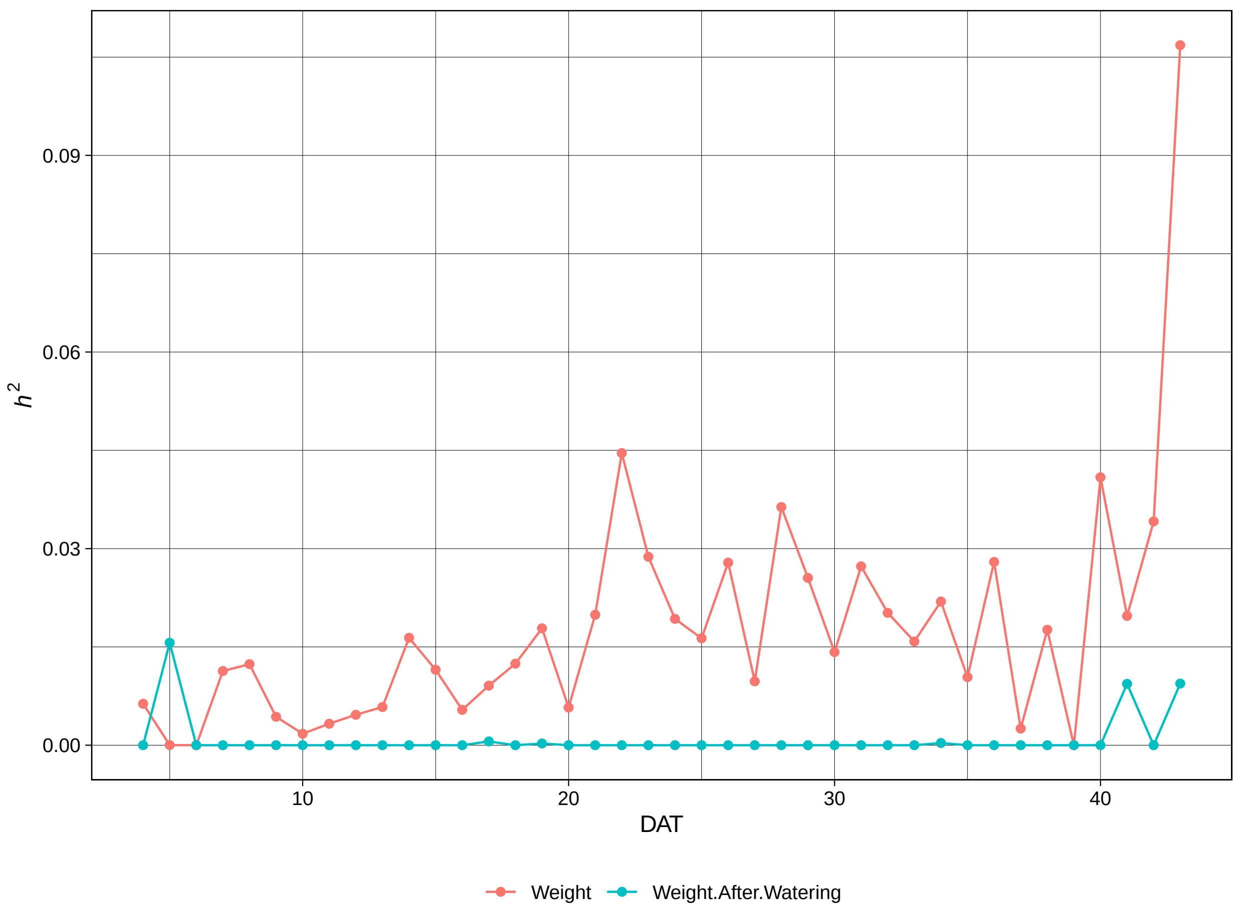

Figure 13: Heritability estimates for SW traits through time.

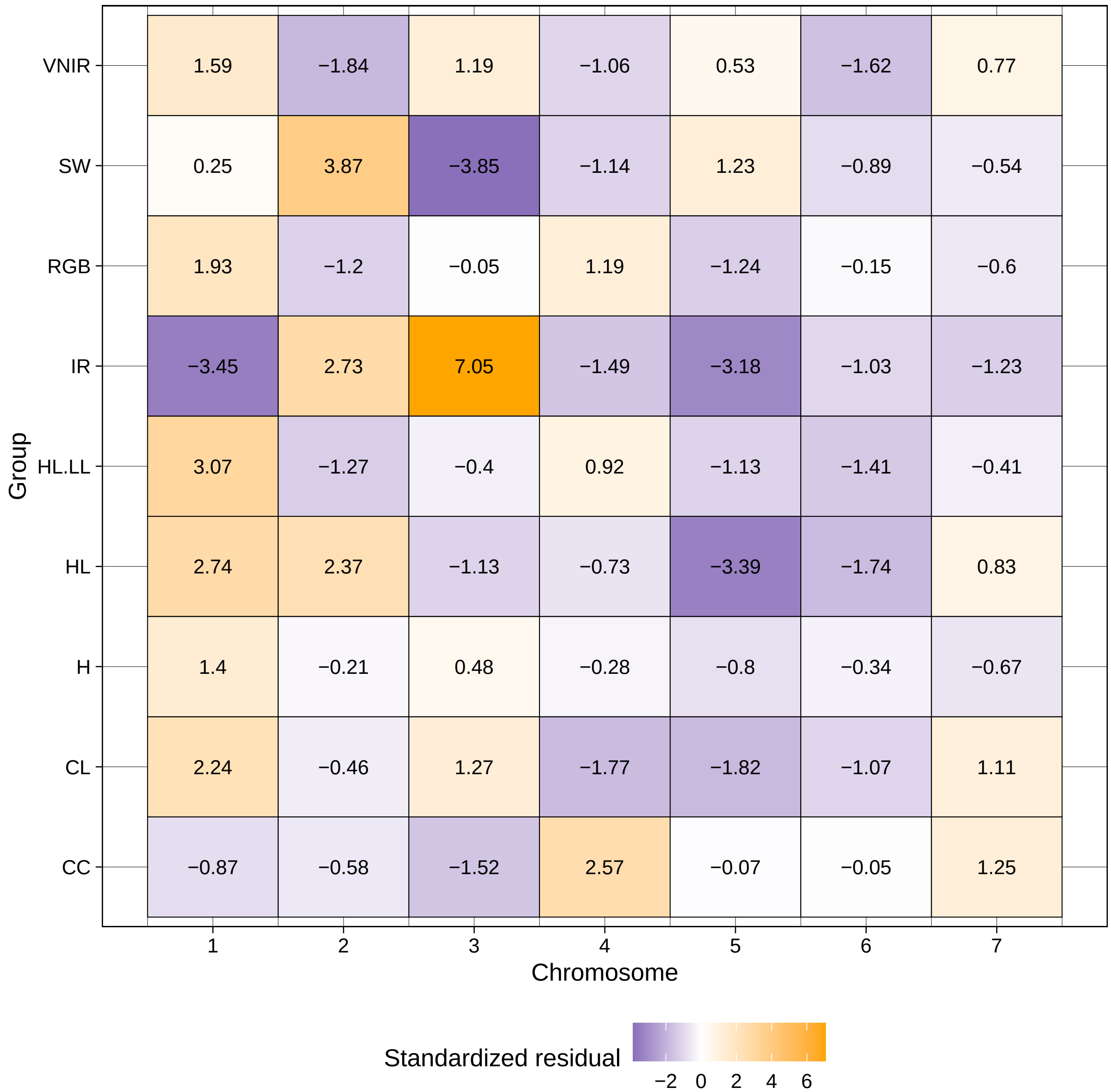

Figure 14: Enrichment of associations per chromosome-group pair Standardized residuals of a chiˆ2-test for homogeneity of the distribution of associated SNP. P-values for chiˆ2-test can be found in STab.3

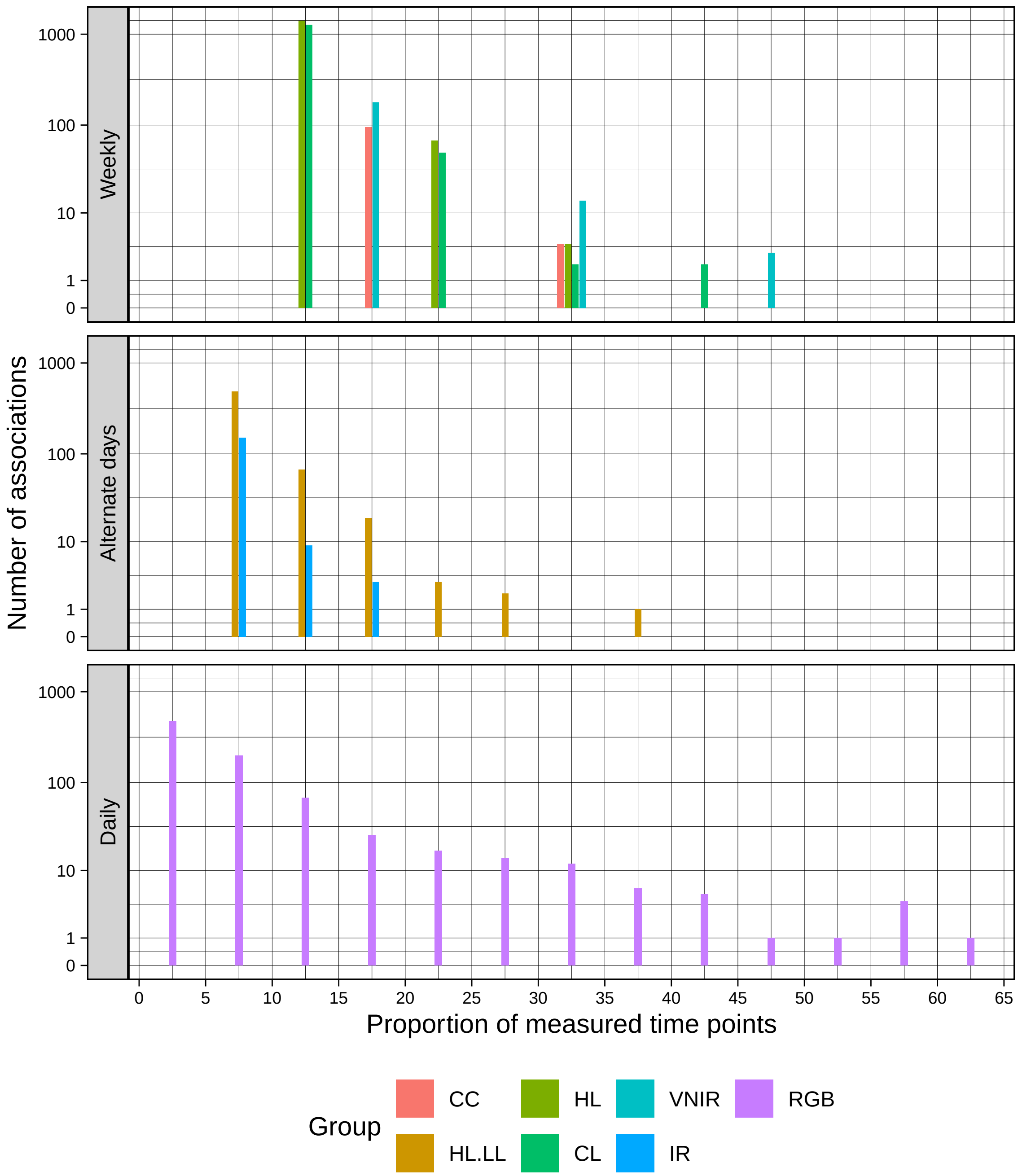

Figure 15: Marker stability for groups with different measurement frequencies.

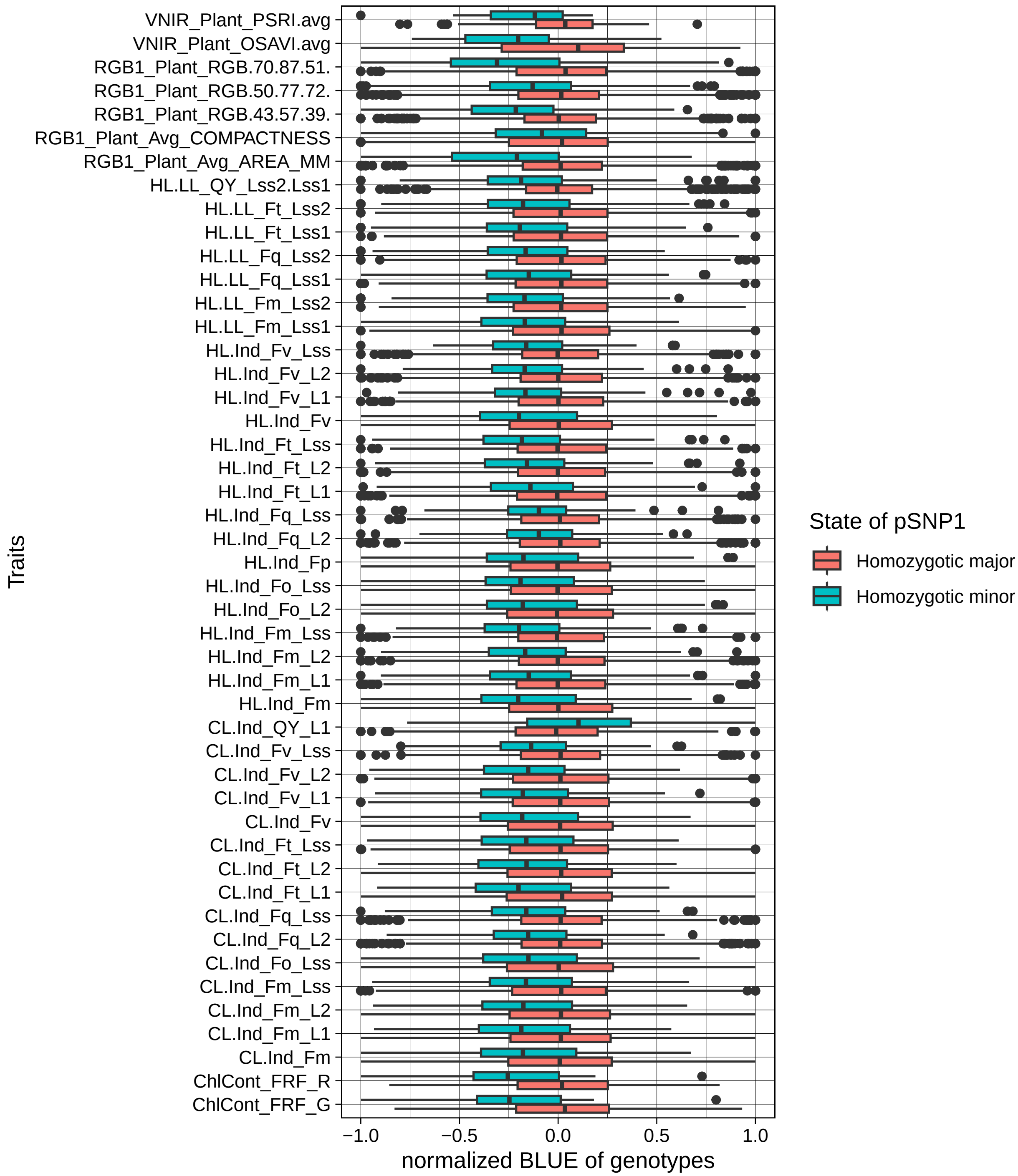

Figure 16: Effect of allele state in pSNP1 Normalized genomic BLUP predictions in associated traits between lines with differing allele states in pSNP1

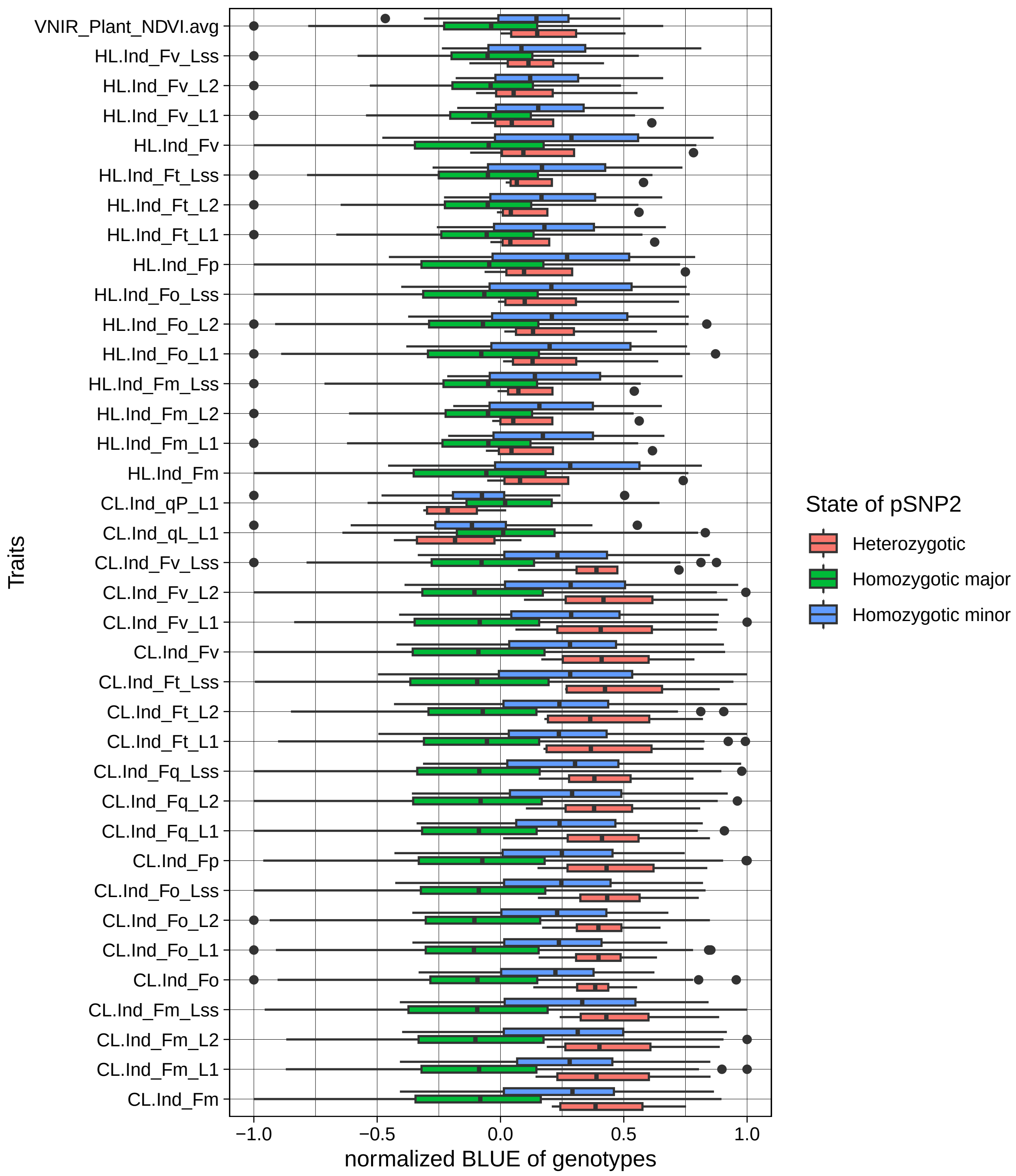

Figure 17: Effect of allele state in pSNP2 Normalized genomic BLUP predictions in associated traits between lines with differing allele states in pSNP2

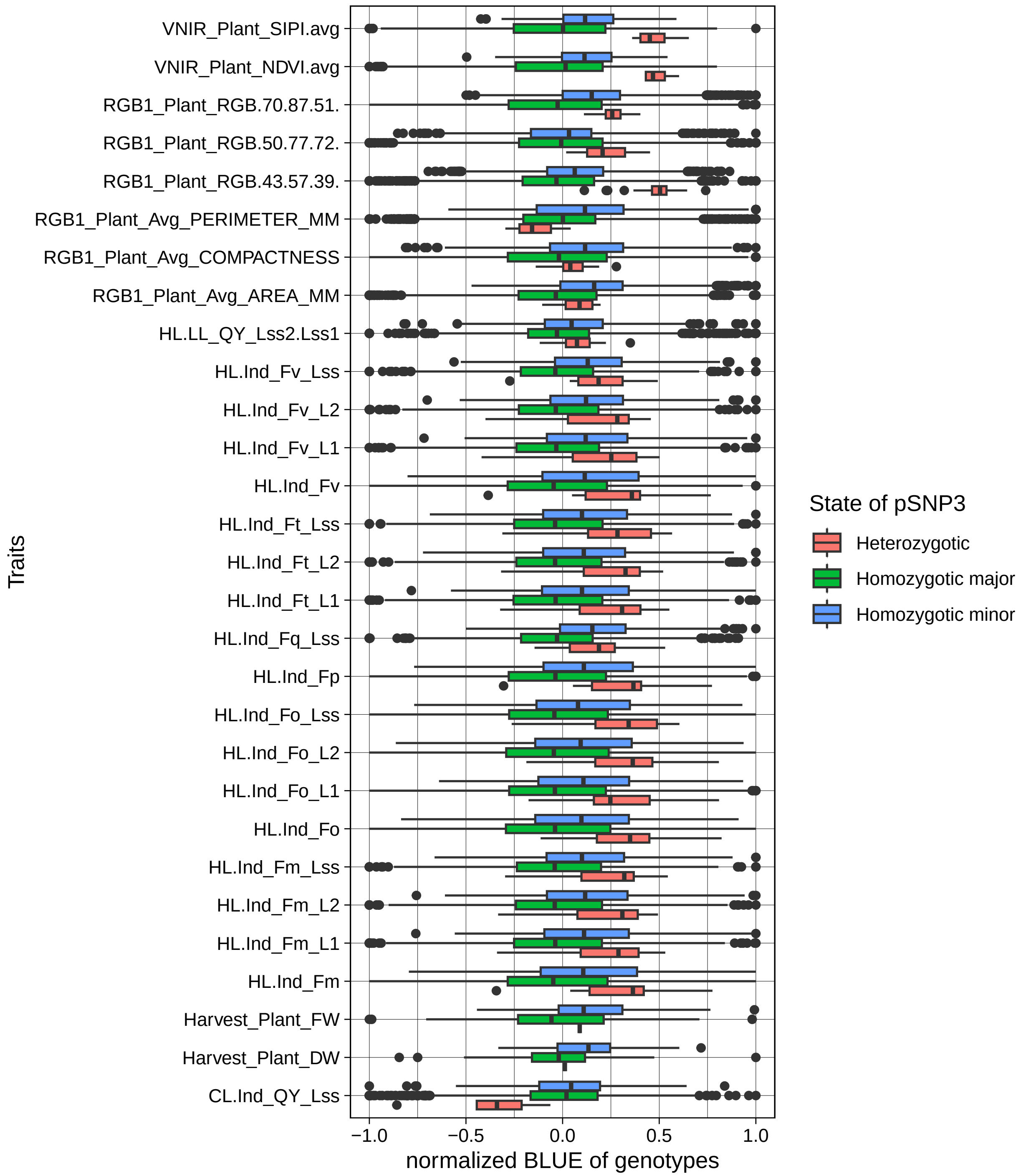

Figure 18: Effect of allele state in pSNP3 Normalized genomic BLUP predictions in associated traits between lines with differing allele states in pSNP3

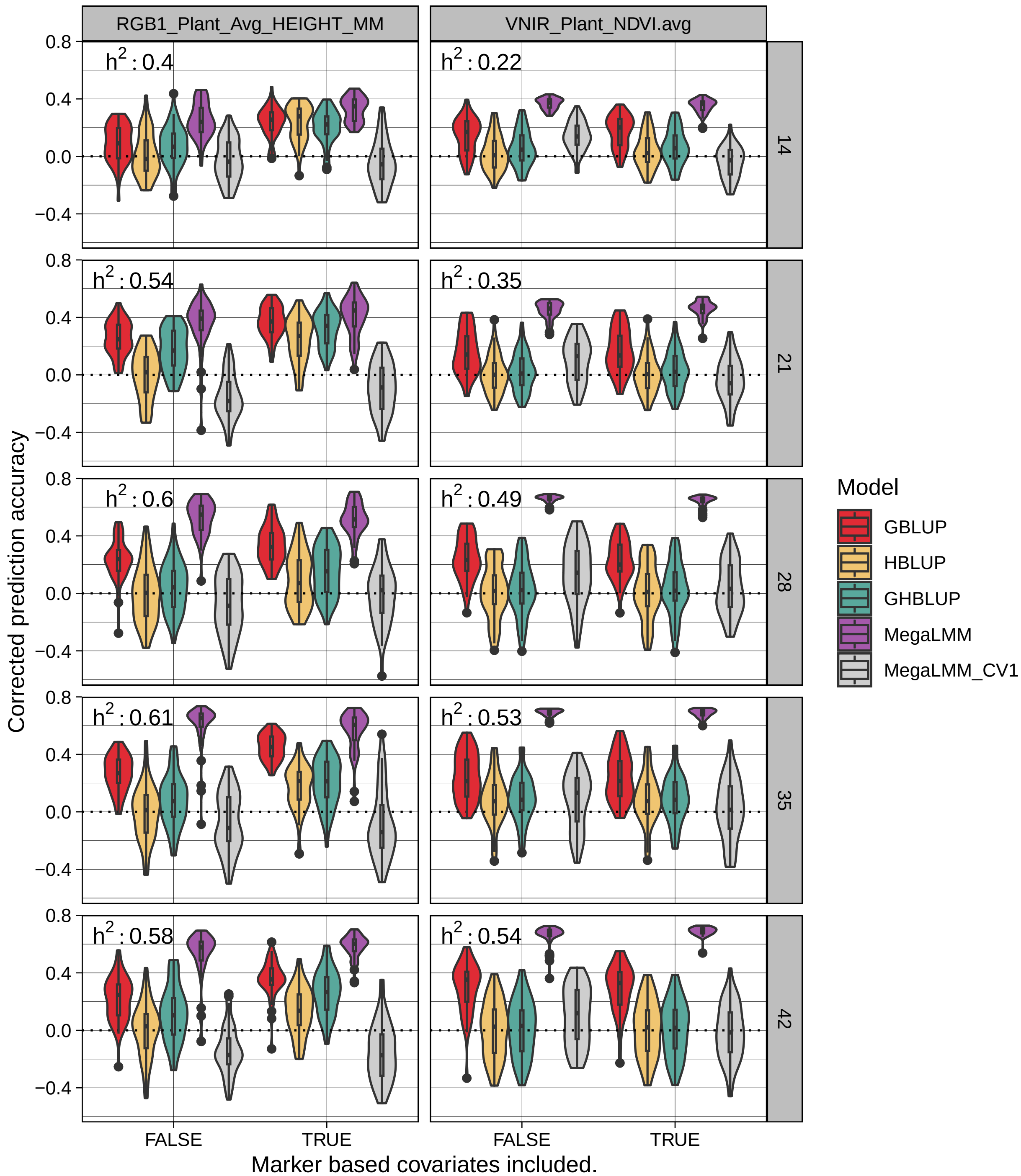

Figure 19: Time specific genomic prediction performance GP accuracy for two yield-related traits at 5 time points.

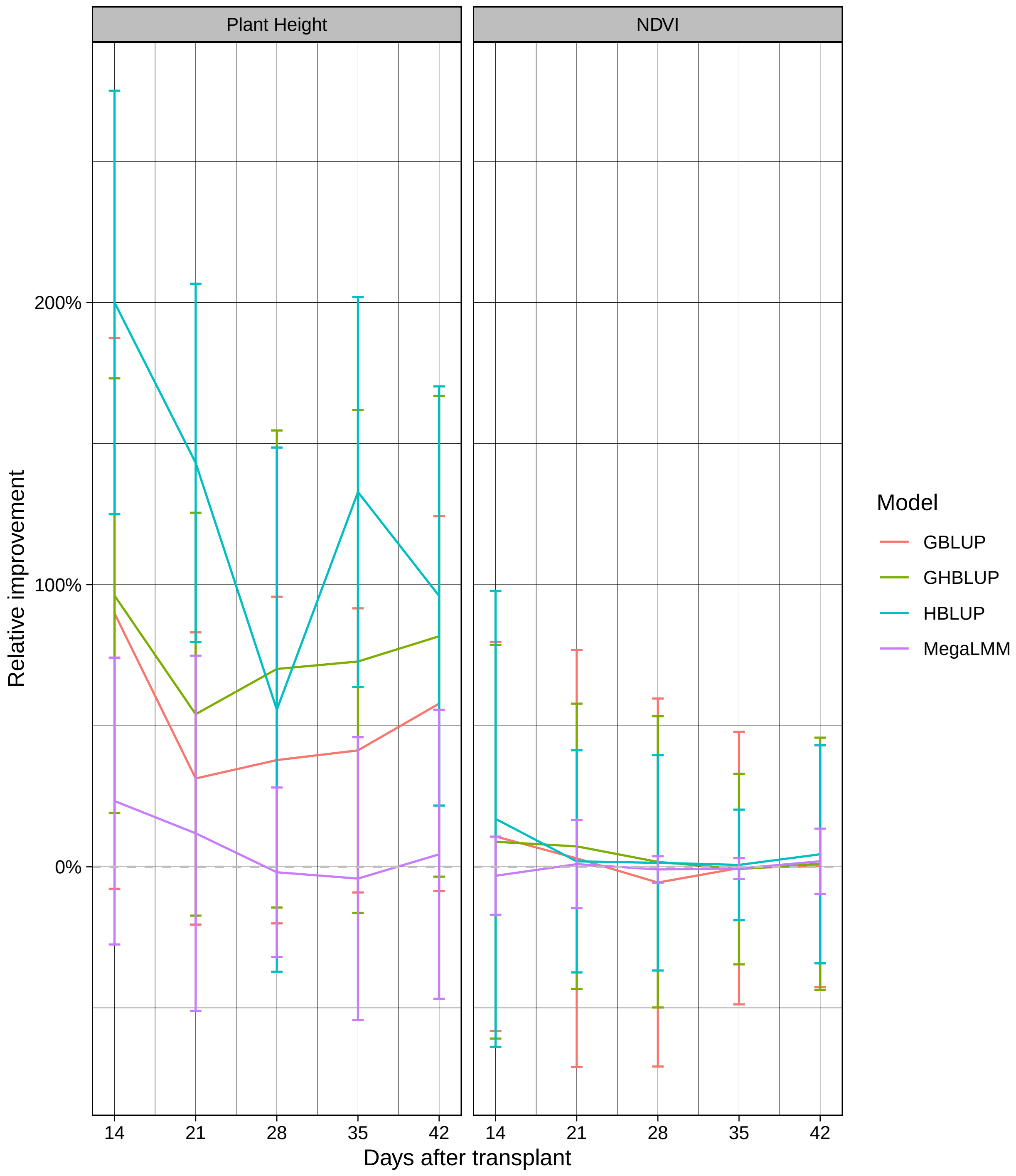

Figure 20: Genomic prediction performance improvement Improvement of prediction accuracy observed after inclusion of marker based covariates. Improvement was calculated as the symmetric mean absolute percentage error and averaged over 50 validation runs.

Table 1: Group assignment (1/2) of all recorded traits. Column names indicate the respective group, while entries are the assigned traits.

HL.LL	HL	CL	RGB
HL.LL_Fm_Lss1	HL.Ind_Fo	CL.Ind_Fo	Avg_AREA_MM
HL.LL_Fm_Lss2	HL.Ind_Fm	CL.Ind_Fm	Avg_PERIMETER_MM
HL.LL_Ft_Lss1	HL.Ind_Fv	CL.Ind_Fv	Avg_COMPACTNESS
HL.LL_Ft_Lss2	HL.Ind_Fp	CL.Ind_Fp	Avg_WIDTH_MM
HL.LL_Fq_Lss1	HL.Ind_Fm_L1	CL.Ind_Fm_L1	Avg_HEIGHT_MM
HL.LL_Fq_Lss2	HL.Ind_Fm_L2	CL.Ind_Fm_L2	RGB.43.57.39.
HL.LL_QY_Lss1	HL.Ind_Fm_Lss	CL.Ind_Fm_Lss	RGB.124.135.89.
HL.LL_QY_Lss2	HL.Ind_Ft_L1	CL.Ind_Ft_L1	RGB.95.112.67.
HL.LL_QY_Lss2.Lss1	HL.Ind_Ft_L2	CL.Ind_Ft_L2	RGB.74.97.81.
	HL.Ind_Ft_Lss	CL.Ind_Ft_Lss	RGB.50.77.72.
	HL.Ind_Fo_L1	CL.Ind_Fo_L1	RGB.70.87.51.
	HL.Ind_Fo_L2	CL.Ind_Fo_L2	
	HL.Ind_Fo_Lss	CL.Ind_Fo_Lss	
	HL.Ind_Fv_L1	CL.Ind_Fv_L1	
	HL.Ind_Fv_L2	CL.Ind_Fv_L2	
	HL.Ind_Fv_Lss	CL.Ind_Fv_Lss	
	HL.Ind_Fq_L1	CL.Ind_Fq_L1	
	HL.Ind_Fq_L2	CL.Ind_Fq_L2	
	HL.Ind_Fq_Lss	CL.Ind_Fq_Lss	
	HL.Ind_QY_max	CL.Ind_QY_max	
	HL.Ind_Fv.Fm_L1	CL.Ind_Fv.Fm_L1	
	HL.Ind_Fv.Fm_L2	CL.Ind_Fv.Fm_L2	
	HL.Ind_Fv.Fm_Lss	CL.Ind_Fv.Fm_Lss	
	HL.Ind_QY_L1	CL.Ind_QY_L1	
	HL.Ind_QY_L2	CL.Ind_QY_L2	
	HL.Ind_QY_Lss	CL.Ind_QY_Lss	
	HL.Ind_NPQ_L1	CL.Ind_NPQ_L1	
	HL.Ind_NPQ_L2	CL.Ind_NPQ_L2	
	HL.Ind_NPQ_Lss	CL.Ind_NPQ_Lss	
	HL.Ind_qN_L1	CL.Ind_qN_L1	
	HL.Ind_qN_L2	CL.Ind_qN_L2	
	HL.Ind_qN_Lss	CL.Ind_qN_Lss	
	HL.Ind_qP_L1	CL.Ind_qP_L1	
	HL.Ind_qP_L2	CL.Ind_qP_L2	
	HL.Ind_qP_Lss	CL.Ind_qP_Lss	
	HL.Ind_qL_L1	CL.Ind_qL_L1	
	HL.Ind_qL_L2	CL.Ind_qL_L2	
	HL.Ind_qL_Lss	CL.Ind_qL_Lss	
	HL.Ind_Rfd_L1	CL.Ind_Rfd_L1	
	HL.Ind_Rfd_L2	CL.Ind_Rfd_L2	
	HL.Ind_Rfd_Lss	CL.Ind_Rfd_Lss	

Table 2: Group assignment(1/2) of all recorded traits. Column names indicate the respective group, while entries are the assigned traits.

CC	VNIR	IR	SW	H
ChlCont_FRF_G	PRI.avg	Temp.avg	Weight	FW
ChlCont_RF_Gcorr	NDVI.avg	deltaT	Weight.After.Watering	DW
ChlCont_SFR_G	PSRI.avg			FWDWratio
ChlCont_FRF_R	OSAVI.avg			
ChlCont_RF_Rcorr	SIPI.avg			
ChlCont_SFR_R	MCARI1.avg			

Table 3: Chi^2-test for homogeneous distribution. Tests the distribution of the set of associated SNP across all chromosomes. The alternative hypothesis is that there is no difference in number of associated SNP between chromosomes.

Trait Group	p-value	Number of associated SNP
IR	3.54e-13	135
RGB	0.321	617
SW	9.97e-05	298
HL.LL	0.0483	337
CC	0.121	77
VNIR	0.12	149
CL	0.039	577
HL	0.000336	661
H	0.82	21

Table 4: Improvement by implementing MFE Paired t-test results of changes in prediction accuracy between models with and without marker fixed effects.

	Trait	DAT	Model	Statistic	p_value
1	RGB1_Plant_Avg_HEIGHT_MM	14	GBLUP	-6.346	7.66e-09
11	RGB1_Plant_Avg_HEIGHT_MM	14	HBLUP	-8.058	2.45e-12
21	RGB1_Plant_Avg_HEIGHT_MM	14	GHBLUP	-5.307	7.49e-07
31	RGB1_Plant_Avg_HEIGHT_MM	14	MegaLMM	-3.816	0.000251
41	RGB1_Plant_Avg_HEIGHT_MM	14	MegaLMM_CV1	0.619	0.538
3	RGB1_Plant_Avg_HEIGHT_MM	21	GBLUP	-5.137	1.44e-06
13	RGB1_Plant_Avg_HEIGHT_MM	21	HBLUP	-7.422	4.32e-11
23	RGB1_Plant_Avg_HEIGHT_MM	21	GHBLUP	-5.442	4.06e-07
33	RGB1_Plant_Avg_HEIGHT_MM	21	MegaLMM	-1.547	0.125
43	RGB1_Plant_Avg_HEIGHT_MM	21	MegaLMM_CV1	-1.889	0.0619
5	RGB1_Plant_Avg_HEIGHT_MM	28	GBLUP	-3.745	0.000306
15	RGB1_Plant_Avg_HEIGHT_MM	28	HBLUP	-2.630	0.00991
25	RGB1_Plant_Avg_HEIGHT_MM	28	GHBLUP	-3.224	0.00172
35	RGB1_Plant_Avg_HEIGHT_MM	28	MegaLMM	0.621	0.536
45	RGB1_Plant_Avg_HEIGHT_MM	28	MegaLMM_CV1	-1.497	0.138
7	RGB1_Plant_Avg_HEIGHT_MM	35	GBLUP	-8.406	6.23e-13
17	RGB1_Plant_Avg_HEIGHT_MM	35	HBLUP	-5.061	2.02e-06
27	RGB1_Plant_Avg_HEIGHT_MM	35	GHBLUP	-3.659	0.000411
37	RGB1_Plant_Avg_HEIGHT_MM	35	MegaLMM	1.262	0.21
47	RGB1_Plant_Avg_HEIGHT_MM	35	MegaLMM_CV1	0.393	0.695
9	RGB1_Plant_Avg_HEIGHT_MM	42	GBLUP	-5.066	2.01e-06
19	RGB1_Plant_Avg_HEIGHT_MM	42	HBLUP	-3.735	0.000318
29	RGB1_Plant_Avg_HEIGHT_MM	42	GHBLUP	-4.301	4.15e-05
39	RGB1_Plant_Avg_HEIGHT_MM	42	MegaLMM	-2.143	0.0357
49	RGB1_Plant_Avg_HEIGHT_MM	42	MegaLMM_CV1	0.276	0.783
2	VNIR_Plant_NDVI.avg	14	GBLUP	-1.169	0.245
12	VNIR_Plant_NDVI.avg	14	HBLUP	-0.686	0.494
22	VNIR_Plant_NDVI.avg	14	GHBLUP	-0.423	0.673
32	VNIR_Plant_NDVI.avg	14	MegaLMM	1.992	0.0493
42	VNIR_Plant_NDVI.avg	14	MegaLMM_CV1	8.603	1.33e-13
4	VNIR_Plant_NDVI.avg	21	GBLUP	-0.319	0.75
14	VNIR_Plant_NDVI.avg	21	HBLUP	-0.086	0.932
24	VNIR_Plant_NDVI.avg	21	GHBLUP	-0.246	0.806
34	VNIR_Plant_NDVI.avg	21	MegaLMM	-0.262	0.794
44	VNIR_Plant_NDVI.avg	21	MegaLMM_CV1	4.401	2.74e-05
6	VNIR_Plant_NDVI.avg	28	GBLUP	0.267	0.79
16	VNIR_Plant_NDVI.avg	28	HBLUP	0.018	0.986
26	VNIR_Plant_NDVI.avg	28	GHBLUP	-0.030	0.976
36	VNIR_Plant_NDVI.avg	28	MegaLMM	2.658	0.00942
46	VNIR_Plant_NDVI.avg	28	MegaLMM_CV1	2.543	0.0126
8	VNIR_Plant_NDVI.avg	35	GBLUP	0.236	0.814
18	VNIR_Plant_NDVI.avg	35	HBLUP	-0.042	0.966
28	VNIR_Plant_NDVI.avg	35	GHBLUP	0.002	0.999
38	VNIR_Plant_NDVI.avg	35	MegaLMM	0.235	0.815
48	VNIR_Plant_NDVI.avg	35	MegaLMM_CV1	1.662	0.0996
10	VNIR_Plant_NDVI.avg	42	GBLUP	0.065	0.948
20	VNIR_Plant_NDVI.avg	42	HBLUP	-0.184	0.855
30	VNIR_Plant_NDVI.avg	42	GHBLUP	0.059	0.953
40	VNIR_Plant_NDVI.avg	42	MegaLMM	-2.760	0.00744

	Trait	DAT	Model	Statistic	p_value
50	VNIR_Plant_NDVI.avg	42	MegaLMM_CV1	2.794	0.00627

Table 5: Fixed effect coefficient averages across models.

Chrom	Pos	Trait	DAT	GBLUP	HBLUP	GHBLUP	MegaLMM	MegaLMM.CV1
1	347256694	NDVI	21	-0.369	-0.174	-0.190	-0.290	-0.566
1	347256694	NDVI	35	-0.539	-0.026	-0.022	-0.644	-0.209
1	347256694	NDVI	42	-0.593	-0.037	-0.054	-0.612	-0.748
1	48317122	NDVI	21	0.214	0.043	0.074	0.177	0.373
2	462299201	Plant height	28	0.606	0.158	0.270	0.660	0.895
2	462299201	Plant height	35	0.551	-0.024	0.040	0.289	1.382
2	56835696	Plant height	14	0.135	0.165	0.119	0.149	-0.074
2	56835854	Plant height	14	0.153	0.154	0.168	0.125	0.172
2	56835854	Plant height	21	0.273	0.284	0.242	0.247	0.520
2	615971733	NDVI	28	-0.252	-0.052	-0.045	-0.296	-0.141
2	615971733	NDVI	35	-0.226	-0.036	-0.039	-0.215	0.057
2	631074842	Plant height	35	0.404	0.228	0.208	0.281	-0.589
2	631074842	Plant height	42	0.392	0.212	0.257	0.369	0.406
2	657275399	NDVI	42	-0.643	-0.040	-0.085	-0.608	-1.073
3	2792228	Plant height	21	-0.199	-0.266	-0.178	-0.177	0.230
3	2792228	Plant height	28	-0.212	-0.175	-0.118	-0.216	-0.256
3	2792228	Plant height	35	-0.267	-0.209	-0.157	-0.347	0.414
3	51800478	NDVI	14	-0.230	-0.016	-0.026	-0.206	-0.304
3	51800478	NDVI	21	-0.126	-0.021	-0.040	-0.145	0.246
3	561359604	Plant height	14	0.204	0.208	0.198	0.190	0.426
3	561359604	Plant height	21	0.257	0.212	0.236	0.258	0.262
3	561359604	Plant height	28	0.218	0.168	0.171	0.228	0.355
3	561359604	Plant height	35	0.226	0.181	0.178	0.164	0.391
3	561359604	Plant height	42	0.237	0.141	0.154	0.301	0.635
3	563926632	NDVI	21	0.402	0.096	0.100	0.458	0.329
3	563926632	NDVI	28	0.514	0.086	0.074	0.601	0.473
4	505022504	Plant height	42	0.341	0.388	0.312	0.352	0.341
4	603676865	Plant height	14	-0.749	-0.741	-0.700	-0.772	-0.769
5	377686445	NDVI	14	-0.320	-0.142	-0.136	-0.347	0.116
5	532872692	Plant height	21	0.338	0.341	0.346	0.324	0.477
5	548720813	NDVI	21	0.377	0.067	0.067	0.295	0.382
7	213626911	Plant height	14	0.184	0.108	0.158	0.135	-0.073
7	213626911	Plant height	42	0.202	0.083	0.158	0.232	-0.234

Supplementary Section 1: GO-enrichment in regions around SNP of interest

pSNP2 For pSNP2, a total of 72 candidate genes were identified within LD-distance. For these, GO terms relating to *ribosome assembly* (GO:0042255, $p = 7.8 \times 10^{-3}$), *protein ubiquination* (GO:0016567, $p = 1.2 \times 10^{-3}$), *translation* (GO:0006412, $p = 5.1 \times 10^{-3}$), and post-transcriptional regulation of expression like *enzyme binding* (GO:0019899, $p = 2.5 \times 10^{-3}$) and *enzyme regulator activity* (GO:0030234, $p = 5.1 \times 10^{-4}$) were significantly enriched. #### pSNP3 For pSNP3 only a single annotatated gene for a zinc finger CCCH domain-containing protein 19-like (*LOC123434274*) was found within LD-decay distance. Considering the two neighboring markers are within LD-decay distance, yet assuming a longer LD-decay than the estimated $\sim 30kb$, we investigated a $1Mb$ genomic region centered on the marker. By doing so, we identified five additional genes: a transmembrane and coiled-coil domain-containing protein 4-like (*LOC123410405*), two serine proteases EDA2 (*LOC123434300*, *LOC123434326*), a CDPK-related kinase 3-like (*LOC123434352*), and a uncharacterized gene (*LOC123434258*), showing no sequence homology to any known genes within the NCBI taxonomy.

ph-1-{1-10} This region, extending from 128.6 Mb to 140.8 Mb on chromosome 1H exceeds the corresponding estimated LD-decay (*i.e.,*$\sim$ 1.29 Mb), indicating either locally increased LD-decay or potential structural variation within the population on chromosome 1H. Within this range there are 3888 candidate genes. In the biological process domain, the four most significantly enriched GO-terms for this set of genes out of a total of 143 enriched terms were all related to biogenesis and cell replication, such as *cellular component organization* (GO:0016043, $p = 4.2 \times 10^{-4}$), *protein-containing complex organization* (GO:0043933, $p = 4.2 \times 10^{-4}$), *microtubule cytoskeleton organization* (GO:0000226, $p = 5.3 \times 10^{-4}$), and nuclear DNA replication (GO:0033260, $p = 7.6 \times 10^{-4}$). Additionally, in the cellular component domain, a broad spectrum of 31 terms were enriched, with *chromosome* (GO:0005694, $p = 9.0 \times 10^{-4}$), its child-terms *chromatin* (GO:0000785, $p = 1.3 \times 10^{-3}$) and *nucleosome* (GO:0000786, $p = 6.3 \times 10^{-4}$), *microtubule organizing center* (GO:0005815, $p = 1.2 \times 10^{-2}$), and *vesicle* (GO:0031982, $p = 1.2 \times 10^{-2}$) among the most significant.

PH early development Additional variants affecting early development were identified on chromosomes 1H (ph-1-11), 2H (ph-3-2), 5H (ph-5-1), and 7H (ph-7-2). For these we consider a LD-window according to their MAF bin for candidate gene inference (see *Materials & Methods*). Candidate genes around these markers showed enrichment in GO terms cell division (GO:0051301, $p = 4.2 \times 10^{-2}$), plant-type cell wall organization or biosynthesis (GO:0071669, $p = 2.7 \times 10^{-3}$), and organ growth (GO:0035265, $p = 4.7 \times 10^{-2}$). Specifically, we identified *LOC123436626* within this set, predicted to encode a cellulose synthase.

ph-2-4 For this marker, the set of genes was significantly enriched in GO terms clustering around *cell wall modification* (GO:0042545, $p = 2.7 \times 10^{-4}$) and *developmental cell growth* (GO:0048588, $p = 1.7 \times 10^{-2}$). Excluding candidate genes of this marker, the set of genes found through associations in the intermediate development period was most strongly enriched for GO terms *cellulose catabolic process* (GO:0030245 , $p = 4.7 \times 10^{-3}$), *intracellular signalling cassette* (GO:0141124 , $p = 1.5 \times 10^{-3}$), and *endoplasmatic reticulum organization* (GO:0007029 , $p = 1.0 \times 10^{-2}$).

Intermediate development For these, GO-terms clustering around *cell wall biogenesis* (GO:0042546, $p = 2.7 \times 10^{-3}$) and *developmental cell growth* (GO:0048588, $p = 4.8 \times 10^{-2}$) were significantly enriched. Additionally, in this developmental stage, plant-plant interactions seem to gain relevance, as several terms clustering around *response to other organism* (GO:0051707, $p = 6.0 \times 10^{-4}$), such as *response to external biotic stimulus* (GO:0043207, $p = 6.0 \times 10^{-4}$) and *defense response* (GO:0006952, $p = 1.2 \times 10^{-2}$) were found enriched.

Supplementary Section 2: MegaLMM parameterization

```r
library(MegaLMM)
run_parameters = MegaLMM_control(
    h2_divisions = 20,
    # Each variance component is allowed to explain between 0% and 100% of the
    # total variation. How many segments should the range [0,100) be divided
    # into for each random effect?
    burn = 1000,
    # number of burn in samples before saving posterior samples.
    thin = 2, #,
    # during sampling, we'll save every 2nd sample to the posterior database.
    K = 100 # number of factors.
)
  Lambda_prior = list(
    sampler = sample_Lambda_prec_horseshoe,
    # function that implements the horseshoe-based Lambda prior
    # described in Runcie et al 2020.
    #See code to see requirements for this function.
    # other options are:
    # ?sample_Lambda_prec_ARD,
    # ?sample_Lambda_prec_BayesC
    prop_0 = 0.1,
    # prior guess at the number of non-zero loadings in the first and most important factor
    delta = list(shape = 3, scale = 1),
    # parameters of the gamma distribution giving the expected change
    # in proportion of non-zero loadings in each consecutive factor
    delta_iterations_factor = 100
    # parameter that affects mixing of the MCMC sampler. This value is generally fine.
  )

  priors = MegaLMM_priors(
    tot_Y_var = list(V = 0.5,   nu = 5),
    # Prior variance of trait residuals after accounting for fixed effects and factors
    # See MCMCglmm for meaning of V and nu
    tot_F_var = list(V = 18/20, nu = 20),
    # Prior variance of factor traits. This is included to improve MCMC mixing,
    # but can be turned off by setting nu very large
    h2_priors_resids_fun = function(h2s,n)  1,
    # Function that returns the prior density for any value of the h2s vector
    # (ie the vector of random effect proportional variances across all random effects.
    # 1 means constant prior.
    # n is the number of h2 divisions above (here=20)
    # 1-n*sum(h2s)/n linearly interpolates between 1 and 0,
    # giving more weight to lower values
    h2_priors_factors_fun = function(h2s,n) 1,
    # See above.
    # sum(h2s) linearly interpolates between 0 and 1,
    # giving more weight to higher values
    # Another choice is one that gives 50% weight to h2==0: ifelse(h2s == 0,n,n/(n-1))
    Lambda_prior = Lambda_prior
    # from above
  )
```